

OPERATING SYSTEMS

WEEK-01 PARTICAL

NAME : P.Yaswanth Surya

ROLL NO: 2520030581

SECTION : 07

>Develop a C program that uses the system calls open(), read(), write(), and close() to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

CODE:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <unistd.h>
```

```
#include <sys/types.h>
```

```
#include <sys/wait.h>
```

```
int main()
```

```
{
```

```
    char command[100];
```

```
    printf("Enter a Linux command: ");
```

```
    scanf("%s", command);
```

```
pid_t pid = fork();

if (pid < 0)
{
    perror("fork failed");
    return 1;
}

if (pid == 0)
{
    // Child process
    printf("\nChild Process\n");
    printf("Child PID: %d\n", getpid());
    printf("Parent PID: %d\n", getppid());

    execlp(command, command, NULL);

    perror("exec failed");
    exit(1);
}
else
{
    // Parent process
    printf("\nParent Process\n");
    printf("Parent PID: %d\n", getpid());
    printf("Child PID: %d\n", pid);

    wait(NULL);
}
```

```

printf("\nChild process completed.\n");

}

return 0;

}

```

```

Error: Unable to fetch some archives, maybe run apt update or try with --fix-missing.
yaswanthp@Yaswanths-MacBook-Air-M1 ~ % nano practical1.c
yaswanthp@Yaswanths-MacBook-Air-M1 ~ % gcc practical1.c
yaswanthp@Yaswanths-MacBook-Air-M1 ~ % ./a.out
Enter a Linux command: ls

Parent Process
Parent PID: 923
Child PID: 924

Child Process
Child PID: 924
Parent PID: 923
#l.c      dir9     execl.c  execl.c  f2       'fork.c .save'  newfile1  part  permission_lab  pipeex.c  private.txt  script.txt
a.out     dir1     exec2.c  execv3.c  f3       ical          o         pipeex.c  practical1.c  public.txt  test.txt
a8        dir9     execl1.c  f1       'fork.c '  linux_task4    open2.c   pipe1.c  practical2.c  script.sh

Child process completed,

```

>Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices.

-uname

```
yaswanthp@Yaswanths-MacBook-Air-M1 ~ % uname -a
Linux Yaswanths-MacBook-Air-M1 6.10.53.12-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
yaswanthp@Yaswanths-MacBook-Air-M1 ~ % uname
Linux
```

-lscpu

```
yaswanthp@Yaswanths-MacBook-Air-M1 ~ % lscpu
---
CPU op-mode(s):    32-bit, 64-bit
Address sizes:     39 bits physical, 48 bits virtual
Byte Order:        Little Endian
CPU(s):            8
On-line CPU(s) list: 0-7
Vendor ID:         GenuineIntel
Model name:        13th Gen Intel(R) Core(TM) i3-1315U
CPU family:        6
Model:             186
Thread(s) per core: 2
Core(s) per socket: 4
Socket(s):         1
Stepping:          3
BogoMIPS:          4991.99
Flags:              fpu_vme_de_pse_tsc_msr_pae_mce_cx8_apic_sep_mtrr_pge_mca_cmov_pat_pse36_clflush_mmx_fxsr_sse_sse2_ss_ht_syscall_nx_pdpelgb_rdtscp_lm_constant_tsc_rep_good_nopl_xtopology_tsc_reliable_nonstop_tsc_cpuid_tsc_known_freq_pni_pclmulqdq_vmx_ssse3_fma_cx16_pcidsse4_1_sse4_2_x2apic_movbe_popcnt_tsc_deadline_timer_aes_xsave_avx_f16c_rdrand_hypervisor_lahf_lm_abm_3dnowprefetch_ssbd_ibrs_i_bpb_stibp_ibrs_enhanced_tpr_shadow_ept_vpid_ept_ad_fsgsbase_tsc_adjust_bmi1_avx2_smep_bmi2_erms_invpcid_rdseed_adx_swap_clflush_opt_clwb_sha_ni_xsaveopt_xsavec_xgetbv1_xsaves_avx_vnni_vnni_umip_waitpkg_gfni_vaes_vpclmulqdq_rdpid_movdiri_movdir64b_fsrw_md_clear_serialize_ibt_flush_lld_arch_capabilities
Virtualization features:
Virtualization:    VT-x
Hypervisor vendor: Microsoft
Virtualization type: full
Caches (sum of all):
L1d:               192 KiB (4 instances)
L1i:               128 KiB (4 instances)
L2:                5 MiB (4 instances)
L3:               10 MiB (1 instance)
NUMA:
NUMA node(s):      1
NUMA node0 CPU(s): 0-7
Vulnerabilities:
Gather data sampling: Not affected
Ghostwrite:        Not affected
Indirect target selection: Not affected
Itlb multihit:     Not affected
L1tf:              Not affected
Mds:               Not affected
Meltdown:          Not affected
Mmio stale data:   Not affected
Old microcode:     Not affected
Reg file data sampling: Mitigation; Clear Register File
Retbleed:          Mitigation; Enhanced IBRS
Spec rstack overflow: Not affected
Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl
Spectre v1:        Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2:        Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S
Srbds:             Not affected
Tsa:               Not affected
Tsx async abort:   Not affected
Vmsync:            Not affected
```

-lsblk

```
yaswanthp@Yaswanths-MacBook-Air-M1 ~ % lsblk
NAME MAJ:MIN RM SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M 1 disk
sdb   8:16   0 159.4M 1 disk
sdc   8:32   0 2G      0 disk [SWAP]
sdd   8:48   0 1T      0 disk /mnt/wslg/distro
```

-ps

```
yaswanthp@Yaswanths-MacBook-Air-M1 ~ % ps
PID TTY          TIME CMD
331 pts/0        00:00:00 bash
991 pts/0        00:00:00 ps
yaswanthp@Yaswanths-MacBook-Air-M1 ~ % ps
```

-top

```

yaswanthp@Yaswanths-MacBook-Air-M1 ~ % top
-----
Tasks: 20 total, 1 running, 23 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 5747.6 total, 5049.1 free, 508.7 used, 332.2 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used, 5238.9 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24236	15292	11500	S	0.0	0.3	0:00.00	systemd
2	root	20	0	3180	2212	2072	S	0.0	0.0	0:00.01	init-systemd(Ub
7	root	20	0	3180	2044	1940	S	0.0	0.0	0:00.00	init
50	root	19	-1	50436	16936	15660	S	0.0	0.3	0:00.22	systemd-journal
85	systemd+	20	0	22416	14524	12032	S	0.0	0.2	0:00.06	systemd-resolve
91	root	20	0	35300	12240	9224	S	0.0	0.2	0:00.55	systemd-udev
111	root	20	0	2888	1948	1820	S	0.0	0.0	0:00.03	chronyd-starter
112	root	20	0	4472	3172	2920	S	0.0	0.1	0:00.00	cron
113	message+	20	0	8820	5400	4716	S	0.0	0.1	0:00.06	dbus-daemon
152	root	20	0	44096	29848	14712	S	0.0	0.5	0:00.22	networkd-dispat
163	root	20	0	18440	9448	8204	S	0.0	0.2	0:00.08	systemd-logind
185	syslog	20	0	220548	5772	4496	S	0.0	0.1	0:00.08	rsyslogd
226	_chrony	20	0	23880	10532	6344	S	0.0	0.2	0:00.12	chrony
247	_chrony	20	0	12096	2456	1816	S	0.0	0.0	0:00.02	chrony
248	root	20	0	5492	2744	2564	S	0.0	0.0	0:00.00	agetty
249	root	20	0	123460	32700	18080	S	0.0	0.6	0:00.16	unattended-upgr
329	root	20	0	3196	1120	980	S	0.0	0.0	0:00.00	SessionLeader
330	root	20	0	3196	1252	1108	S	0.0	0.0	0:00.10	Relay(331)
331	khyat_x+	20	0	6268	5520	3848	S	0.0	0.1	0:00.24	bash
332	root	20	0	8644	5592	4752	S	0.0	0.1	0:00.01	login
376	khyat_x+	20	0	22752	13388	10936	S	0.0	0.2	0:00.17	systemd
378	khyat_x+	20	0	22920	4132	2128	S	0.0	0.1	0:00.00	(sd-pam)
407	khyat_x+	20	0	6136	5420	3864	S	0.0	0.1	0:00.05	bash
1001	khyat_x+	20	0	8276	6052	3900	R	0.0	0.1	0:00.00	top